

GRACE DOUGLAS

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I am a human-subjects researcher that builds human-like models to describe road user behavior in shared urban spaces. Specifically, my dissertation work explores cross-cultural representations of vulnerable road users in mixed-traffic driving environments, aiming to develop fair, safe, and shareable representations of real-time mixed-traffic negotiations.

EDUCATION

New York University – NYU | Ph.D. Candidate

Expected: May 2027

Transportation Systems, Civil & Urban Engineering, Advisor: Dr. Linda Ng Boyle, Vice Dean of Research Tandon SOE

Keywords: human factors, multimodal transportation, human-automation interaction, driver safety, statistical modeling, vehicle-pedestrian interaction, driver behavior modeling

University of Washington – UW | Master of Science

March 2023

Industrial & Systems Engineering | GPA: 3.7

Relevant Coursework: Stochastic Processes in Engineering, Linear Optimization, Statistical Quality Engineering, Robust Design for Process Improvement, Human Performance Modeling, Engineering Simulation, Inferential Data Analysis

University of Wisconsin – UW - Madison | Bachelor of Science

May 2021

Industrial & Systems Engineering, Statistics | GPA: 3.4

Relevant Coursework: Statistical Theory, Statistics in Engineering, Statistical Experimental Design, Industrial Data Analytics, Operations Research, Intro to Optimization, Intro to Human Factors and Design, Probabilistic Modeling

RESEARCH

Human Factors & Urban Ergonomics Lab (HumanFUEL) – NYU

September 2023 - Present

Advised by Dr. Linda Ng Boyle | Vice Dean of Research, NYU Tandon SOE - C2SMART Institute

1. Virtual Worlds for Real Agents: Validating VR Pedestrian Behavior | Dissertation Research

- Validating VR as a testbed by comparing VR walking to naturalistic pedestrian traces at matched real-world sites.
- Multi-site collaboration with Cornell Tech; tests presence, workload, and motion sickness against ecological validity.

2. Eyes Up! Egocentric Collections of Pedestrian Decision-Making | Dissertation Research | C2SMART

- Instrumented NYC pedestrians with Meta Aria Gen 1 (gaze, SLAM, IMU, GPS), heart rate, and time-aligned think-aloud.
- Models gaze, walking speed, and traffic visibility against per-step isovist geometry built from real-world SLAM.
- Within-subjects goal-directed vs. exploratory walks, ~20 participants on routes they already know.

3. So Pedestrian: Detection Displays in Automated Vehicles | NSF Collaboration with UT Austin

- Built a Unity SAE Level 4 simulator with 138° tri-view display and five levels of detection detail (LOD0–LOD4).
- Tests whether displays *augment* or *displace* passenger awareness, measured on unflagged ambient pedestrians.
- Across 488 encounters (27 participants), moving pedestrians were detected 90–98% vs. 44% when occluded at 7 m.
- Added display detail raised trust without raising workload but never recovered the miss. Accepted to AutoUI 2026.

4. Pedestrian Exposure for Crash Prediction | National Highway Traffic Safety Administration (NHTSA)

- Built intersection-level ML models of pedestrian crash risk from standardized physical-activity exposure measures.
- Misclassification framework over 13,706 Seattle intersections isolating systematic model failure from random error.
- Concentrated 6.6% of total error into 1.03% of locations; street-view audits recovered three omitted feature classes.

5. The Next Mobile Office: How will we Work in Self-Driving Cars? | National Science Foundation (NSF)

- Measuring takeover performance after virtual-meeting engagement in conditionally automated vehicles.
- Emergency vs. Planned takeover treatments randomly assigned during active meeting engagement.

Lakdawala Lab – Emory University School of Medicine

[Jan 2026] – Present

Research Scientist, Computer Vision | PI: Dr. Seema Lakdawala

- Pose estimation and identity tracking for 5–6 dye-marked ferrets under dense occlusion in influenza-transmission studies.
- Decoupled identity from pose with SLEAP creator Dr. Talmo Pereira: Random Forest HSV classifier, Hungarian matching.
- Confidence- and margin-gated assignment abstains rather than guessing, so occlusion swaps cannot propagate.
- Architected the AWS GPU pipeline (EC2/S3, sleap-nn/PyTorch/CUDA): ~1.5 h per 2 h video, streaming at flat memory.

Human Factors and Statistical Modeling Lab – UW

June 2021 – August 2023

Advised by Dr. Linda Ng Boyle | Professor, Chair, UW Industrial & Systems Engineering

1. Cross-Cultural Mapping of Older Drivers | National Science Foundation (NSF)

- This study compared SHRP2 NDS database and Survey of Health, Aging and Retirement in Europe (SHARE) to identify common lifestyle, health, and personality characteristics to profile driving performance across cultural boundaries.
2. **Multimodal Environments & Multitasking Driving Behaviors** | *Federal Highway Administration (FHWA)*
 - Used the largest collected naturalistic driving database (SHRP2 NDS) to examine in-vehicle driving behavior.
 - Applied Dynamic Bayesian Networks to investigate dependencies between dynamic and static contextual predictors in driver behavior modeling.
 - Derived a distracted driver behavior typology from 39,271 one-second observations of naturalistic crosswalk traversals using a Hidden Markov Model and Gaussian Mixture Model pipeline, identifying regulatory responsiveness (not distraction frequency) as the safety-critical dimension.
 3. **Measuring Pedestrian Exposure Using Electronic Devices** | *National Highway Traffic Safety Administration (NHTSA)*
 - Examined pedestrian exposure and risk to collision in Seattle, WA by analyzing spatial-temporal data of pedestrian walking behavior.
 - Implemented zero-inflated count models to predict pedestrian risk dependent on environmental predictors.

Cognitive Systems Lab – UW-Madison

May 2017 – August 2019

Advised by Dr. John Lee | Professor, UW-Madison Industrial & Systems Engineering

- Conducted an ethnographic study recording driver short-stop behavior as a communication tool in uncontrolled multimodal environments.
- Labeled vehicle-pedestrian interaction with ANVIL video coding software, modeled proxemics and kinesics cues in R.
- Utilized NADs miniSim software for naturalistic driving environments to validate short-stop vehicle deceleration profiles.

PUBLICATIONS AND PRESENTATIONS

Peer-Reviewed Journal Articles

Douglas, G., Pereira, T. D., & Lakdawala, S. S. (in preparation). *Decoupling pose from identity in multi-animal pose tracking*. PLOS Biology (protocol paper).

Douglas, G., & Boyle, L. N. (2026, accepted). *A Distracted Driver Behavior Typology: The Role of Regulatory Environments*. Accident Analysis & Prevention.

Douglas, G., Mooney, S. J., Prendez, D. M., & Boyle, L. N. (2026). *A Misclassification Framework for Identifying Systematic Error in Pedestrian Crash Prediction: Evidence from Seattle, WA*. Accident Analysis & Prevention, 235, 108633. [link]

Li, J., Douglas, G., MacKenzie, D., & Boyle, L. (2024). *Examining drivers' secondary task engagement at intersections using naturalistic driving data*. Available at SSRN 5027098. [link]

Domeyer, J., Dinparastdjadid, A., Lee, J. D., Douglas, G., Alsaid, A., & Price, M. (2019). *Proxemics and Kinesics in Automated Vehicle–Pedestrian Communication: Representing Ethnographic Observations*. Transportation Research Record, 2673(10), 70–81. [link]

Conference Proceedings, Technical Reports & Presentations

Douglas, G. A., & Boyle, L. N. (2026). *So Pedestrian: Designing AV Detection Displays Around Human Perception*. Adjunct Proceedings of the 18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26), Gothenburg, Sweden. [link]

Douglas, G., Mooney, S. J., Prendez, D. M., & Boyle, L. N. (2026). *Systematic Misclassification in Pedestrian Crash Modeling*. Poster presented at the Joint Statistical Meetings (JSM), Boston, MA.

Douglas, G. A., & Boyle, L. N. (2025). *Virtual worlds for real agents: Validating virtual pedestrian behavior for vehicle assistance systems*. Adjunct Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications, 73–77. [link]

Douglas, G., Boyle, L. N., Mooney, S., Sealens, B., Moudon, A., & Ebel, B. (2025). *Pedestrian Exposure Measures for Different Scales of Analysis*. Paper presented at the 104th Annual Meeting of the Transportation Research Board, Washington, D.C.

Douglas, G., Li, J., Khaloei, M., Liu, J., Guo, H., Zou, T., Boyle, L. N., MacKenzie, D. (202X, pre-print). *Investigating How Multimodal Environments Affect Multitasking Driving Behaviors* (Report No. DOT HS xxx xxx). Federal Highway Administration. [link]

Douglas, G., Boyle, L. N., Saelens, B., Mooney, S., Moudon, A. V., Ebel, B. E., & Kim, H. (2024). *A framework for assessing pedestrian exposure using GPS and accelerometer data* (Report No. DOT HS 813 583). National Highway Traffic Safety Administration. [link]

Douglas, G., Boyle, L. N., Kim, H., Moudon, A., Mooney, S., Saelens, B., & Ebel, B. (2022). *A Framework to Assess Pedestrian Exposure Using Personal Device Data*. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 66(1), 320–324. [link]

CERTIFICATIONS

Internal Review Board (IRB): CITI Program for Human Subjects Research

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Matplotlib, Seaborn), C#, SQL, R (tidyverse, ggplot2, lme4, lmerTest), MATLAB, Bash/Zsh, $\text{\LaTeX}$

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, Deep Learning (CNNs, Transfer Learning), Reinforcement Learning (PPO, Imitation Learning, Behavior Cloning), Ensemble Methods (XGBoost, Random Forest), Latent State Models (Hidden Markov Models, Gaussian Mixture Models), Supervised/Unsupervised Learning, Classification (behavior, gaze, appearance), Confidence Calibration & Abstention/Rejection Thresholding, Human-in-the-Loop (HITL) Model Auditing, Model Training & Fine-tuning, Predictive Modeling

Computer Vision: OpenCV, SLEAP / sleap-nn (multi-animal pose estimation, top-down centroid + centered-instance models, identity assignment under occlusion), Hungarian / Optimal Assignment, Eye-tracking Analysis (gaze metrics, attention classification), Image Processing

XR Development & Optimization: Unity (OpenXR, Netcode, XR Interaction Toolkit, SteamVR), Unreal Engine, Android VR Deployment (ADB, SideQuest), Real-time Profiling (LOD, Occlusion Culling, Instanced Materials, Draw Call Batching), Multiplayer Networked Systems

Statistics & Research: Mixed-effects Modeling (lme4, lmerTest), Bayesian Inference (MCMC, Dynamic Bayesian Networks), Spatial Statistics (Local Moran's I / LISA, spatial-temporal validation), Zero-inflated & Count Models, Experimental Design (within/between-subjects, factorial ANOVA/MANOVA, Williams sequences), Monte Carlo & Spatial-Temporal Cross-Validation, Physiological Signal Processing (ECG, heart rate), Isovist & Visibility Analysis, Standardized Instruments (NASA-TLX, SBSOD, BART), Think-Aloud Protocol & Video Coding (ANVIL), Human-Computer Interaction (HCI)

Data Visualization & Analysis: Matplotlib, Seaborn, ggplot2, Vega-Lite, Rerun, QGIS, Jupyter Notebooks

Simulation & Modeling: SUMO, Flow, miniSim (ISAT), GAMS, AMPL, Blender, MotionBuilder, MOTIVE

Infrastructure & DevOps: Git/GitHub (LFS), Docker/Singularity, SLURM (HPC - NYU Greene/Torch), AWS (GPU EC2, S3, IAM), CUDA/GPU Inference, Streaming & Batch Inference Pipelines, Conda, FFmpeg, API Integration, Unix/Linux Environment

Hardware & Sensors: Optitrack, Sony Mocopi, Meta Quest Pro, Meta Project Aria Glasses (Gen 1), Pupil Labs, Garmin HRM-Dual / COROS HR, NADS MiniSim Quarter Cab, Custom Embedded Systems, Sensor Fusion, Real-time Data Synchronization

LEADERSHIP & INVOLVEMENT

New York University, Establishing HumanFUEL | Brooklyn, NY

January 2024 – Present

Founding research lab member

- Order, install, and calibrate new lab equipment: 2 vehicle simulators, 2 free-motion simulators, eye-tracking equipment, motion capture systems, LANs for real-time synchronization.
- Design lab space for human-subjects experiments, develop lab name and logo, and create lab space use documentation (data storage on secure servers, procedures for single- and multiplayer experimentation, version control etiquette for collaborations).
- Coordinate with building managers, administration, technology manufacturers and equipment suppliers.
- Interview potential undergraduate and graduate candidates for research lab positions.

University of Washington, HFES Student Chapter | Seattle, WA

November 2021 – August 2023

Communications Director

- Maintain UW HFES Chapter website.
- Communicate with industry professionals to connect university students with future professional opportunities.
- Develop outreach surveys and Q&A sessions to encourage student body membership.

University of Wisconsin – Madison, Women's Varsity Soccer | Madison, WI

August 2016 - 2021

NCAA Division 1 Student-Athlete | Team Captain

- 4-year Varsity Letter winner; Athletic Scholarship Recipient.
- Won Big Ten Conference Title (2019) and contributed to team appearances in NCAA Sweet 16 (2018, 2019).
- One of two Wisconsin student athletes chosen for the Big Ten COVID-19 'Return to Play' Taskforce.

INDUSTRY

Seagate Technology | Minneapolis, MN

June 2020 – August 2021

Wafer Manufacturing and Systems Processing Engineering Intern

- Aided in successfully implementing a new tool set for operator use in the largest disk drive manufacturing facility.
- Used system software APF/AMA to analyze current usage of non-production wafers (memory storage disk inside of a disk drive).
- Optimized the use of non-production 'wafers' to satisfy capacity demands of newly implemented machines.

ICR, Inc. | Washington, D.C.

May 2019 - July 2019

Data Science Intern

- Preprocessed real-time data streams in Python.
- Formulated bash scripts in Ubuntu systems for lengthy programs.
- Researched and modeled algorithms for neural networks used by DOD agencies to identify anomalous events.